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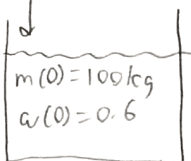
CHE 210, Midterm 2

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1)

$$\dot{m}_i = 10 \text{ kg/min}$$

$$\omega_i = 0.1$$



$$k = 0.15 \text{ min}^{-1}$$

$$\dot{m}_e = k m_{sys}, \quad \dot{m}_e(0) = 15 \text{ kg/min}$$

$$\omega_e = \omega(t)$$

a) Find $m_{sys}(t)$

$$\frac{dm_{sys}}{dt} = \dot{m}_i - \dot{m}_e = 10 - 0.15 m$$

$$\frac{dm_{sys}}{dt} = 10 \frac{\text{kg}}{\text{min}} - 0.15 \frac{1}{\text{min}} \cdot m$$

$$\frac{dm_{sys}}{dt} + 0.15 m = 10$$

$$w(t) = e^{\int 0.15 dt} = e^{0.15 t}$$

Integrating factor

$$e^{0.15 t} \frac{dm}{dt} + 0.15 e^{0.15 t} m = 10 e^{0.15 t}$$

$$\int \frac{d}{dt} (e^{0.15 t} m) = \int 10 e^{0.15 t}$$

$$e^{0.15 t} m = 10 \frac{1}{0.15} e^{0.15 t} + C$$

$$m_{sys}(t) = 66.7 + C e^{-0.15 t}$$

$$m_{sys}(0) = 100 = 66.7 + C$$

$$C = 33.3$$

$$m_{sys}(t) = 66.7 + 33.3 e^{-0.15 t}$$

b) $m_{sys}(10) = 66.7 + 33.3 e^{-0.15(10)} = 74.13 \text{ kg}$

$$\begin{array}{r} 1 \overline{) 45} \\ 2 \overline{) 49} \\ \hline 1194 \end{array}$$

same fancy math

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$$1c) \quad \frac{dm_A}{dt} = \dot{m}_{A,in} - \dot{m}_{A,out}$$

$$\left| \begin{aligned} \frac{dm_A}{dt} &= \frac{d(\omega m_{s+r})}{dt} \\ \dot{m}_{A,in} &= 0.1 \cdot (10) = 1 \\ \dot{m}_{A,out} &= k m_A = k(\omega m) \end{aligned} \right.$$

$$\frac{d(\omega m)}{dt} = 1 - 0.15(\omega m) \leftarrow \text{This is separable!}$$

$$m \frac{d\omega}{dt} + \omega \frac{dm}{dt} = 1 - 0.15 \omega m \leftarrow \text{why? you are making this more difficult}$$

$$\frac{dm}{dt} = 10 - 0.15 m$$

$$m \frac{d\omega}{dt} = 1 - 0.15 \omega m - \omega(10 - 0.15 m)$$

$$m \frac{d\omega}{dt} = 1 - 0.15 \omega m - 10\omega + 0.15 m \omega$$

$$m \frac{d\omega}{dt} = 1 - 10\omega$$

$$\frac{d\omega}{dt} = \frac{1 - 10\omega}{m} = \frac{1 - 10\omega}{66.7 - 33.3 e^{-0.15t}}$$

multiply by $\frac{e^{0.15t}}{e^{0.15t}}$

$$\int \frac{d\omega}{1 - 10\omega} = \int \frac{1}{66.7 - 33.3 e^{-0.15t}} d\omega = \int \frac{-10}{66.7 - 33.3 e^{-0.15t}} d\omega$$

$$\left. \begin{aligned} \omega &= 1 - 10\omega \\ d\omega &= -10 d\omega \\ \downarrow d\omega &= \frac{d\omega}{-10} \end{aligned} \right\}$$

$$= -10 \int \frac{e^{0.15t}}{+333 - 667 e^{0.15t}} d\omega$$

$$x = +333 - 667 e^{0.15t}, \quad dx = -100.05 e^{0.15t} d\omega$$

$$d\omega = \frac{dx}{-100.05 e^{0.15t}}$$

$$-\frac{1}{10} \int \frac{1}{\omega} d\omega$$

$$= -\frac{1}{10} \ln(1 - 10\omega)$$

$$= -10 \int \frac{e^{0.15t}}{x} \cdot \frac{dx}{-100.05 e^{0.15t}}$$

you did some good work here
just unnecessarily hard

$$= \frac{10}{100.05} \int \frac{1}{x} dx$$

$$= \frac{10}{100.05} \ln |333 - 667 e^{0.15t}|$$

$$-\frac{1}{10} \ln(1 - \omega) = 0.1 \ln |333 - 667 e^{0.15t}| + C$$

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$$1c) \quad -\frac{1}{10} \ln(1-\omega) = 0.1 \ln(333 - 667 e^{0.15t}) + C$$

$$\omega(0) = 0.6$$

$$-\frac{1}{10} \ln(0.4) = 0.1 \ln(333 - 667 e^0) + C$$

$$-0.0916 = 0.1 \cdot (5.811) + C$$

$$C = -0.673$$

$$-\frac{1}{10} \ln(1-\omega) = 0.1 \ln(333 - 667 e^{0.15t}) - 0.673$$

Implicit solution

$$t = 10 \text{ min, } \omega(t) = ?$$

$$-\frac{1}{10} \ln(1-\omega) = 0.1 \ln(333 - 667 e^{0.15(10)}) - 0.673$$

$$-\frac{1}{10} \ln(1-\omega) = 0.1 (2.656.3) - 0.673$$

$$\ln(1-\omega) = -2649.55$$

$$(1-\omega) = e^{-2649.55}$$

$$\omega = 1 - e^{-2649.55}$$



Overall Balance

$$B_2 = \frac{150}{0.25} = 600$$

2) reactor balance

$$A: A_3 = A_2 - 2F_1 = 2160$$

$$B: B_3 = B_2 - F_1 - F_2 = 450$$

$$P: P_3 = 0 + F_1 - F_2 = 90$$

$$Q: Q_3 = Q_2 + F_2 = 6450 + 30 = 6480$$

splitter balance

$$A: A_4 = A_3 + A_6 = 2160$$

$$Y = \frac{A_5}{A_4} = \frac{B_5}{B_4} = \frac{Q_5}{Q_4}$$

$$\frac{10}{2160} = \frac{2}{B_4} = \frac{30}{Q_4}$$

$$B_4 = 432, Q_4 = 6480$$

$$Q_6 = Q_4 - Q_5 = 6450$$

$$B_6 = B_4 - B_5 = 430$$

Mix balance

$$A: A_2 = A_1 + A_6 \Rightarrow A_6 = 2400 - 250 = 2150$$

$$B: B_2 = B_1 + B_9 + B_6 \Rightarrow B_9 = B_2 - B_1 - B_6$$

$$Q: Q_2 = Q_6 = 6450$$

$$B_9 = 600 - 152 - 430 = 18$$

Top separator

$$B: B_7 = B_9 = 18$$

$$Q: Q_7 = Q_8 = 1$$

2b) Single pass

- selectivity B

$$S_{B \rightarrow P} = \frac{F_1 - F_2}{F_1 + F_2} = \frac{120 - 30}{120 + 30} = \frac{90}{150} = 0.6 = 60\%$$

- yield B $\rightarrow$ P

$$Y_{B \rightarrow P} = \frac{F_1 - F_2}{B_2} = \frac{120 - 30}{600} = \frac{90}{600} = 0.15 = 15\%$$

2b) overall

selectivity

$$S_{B \rightarrow P}^{ov} = \frac{f_1 - f_2}{f_1 + f_2} = \frac{90}{150} = 0.6 = 60\%$$

yield

$$Y_{B \rightarrow P}^{ov} = \frac{f_1 - f_2}{B_1} = \frac{90}{152} = 0.59 = 59\%$$

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